

# IPRECON 2026

## IEEE INTERNATIONAL POWER AND RENEWABLE ENERGY CONFERENCE

College of Engineering Karunagappally, Kerala, India • 4 - 6 September 2026

### TRACK SCHEDULE

**Notes:** Each oral presentation is allotted 15 minutes (12 minutes presentation + 3 minutes Q&A). Sessions marked **(Parallel)** run simultaneously in separate halls. Authors are requested to report to their session hall at least 15 minutes before the session begins. Virtual presenters will receive the online meeting link by e-mail on their registered address. For any clarification regarding your session, please contact the organising committee.

#### Day 1 - Friday, 04 September 2026

##### Track 1 - Renewable Energy & Storage - Session I (Parallel)

10:45 AM - 12:30 PM • 6 presentations × 15 min • in parallel with Track 2

| PAPER ID | PAPER TITLE                                                                                                                                                         | PRESENTATION TIME   |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| 3        | Utilizing Human Locomotion to Harvest Piezo Electric Energy                                                                                                         | 10:45 AM - 11:00 AM |
| 4        | The Correlative Study of Diverse Fuel Cell Technologies                                                                                                             | 11:00 AM - 11:15 AM |
| 275      | A new Hybrid Perturb & Observe-Tianji's Horse Racing Optimization Algorithm for Enhancing PV Power Extraction Under Partial Shading Scenarios                       | 11:15 AM - 11:30 AM |
| 302      | EV WIREFREE- Smart Dynamic Wireless Charging System for Electric Vehicles with Embedded AI and Gyro Sensors                                                         | 11:30 AM - 11:45 AM |
| 316      | Automated I-V Tracer Circuit for Inspection of Bypass Diodes in Photovoltaic Module                                                                                 | 11:45 AM - 12:00 PM |
| 355      | Techno-Economic and Simulation-Based Assessment of Energy Storage Systems for Residential and Utility-Scale Solar Applications: A Case Study of Jalgaon City, India | 12:00 PM - 12:15 PM |

##### Track 2 - Computing, AI & Data Science - Session I (Parallel)

10:45 AM - 12:30 PM • 6 presentations × 15 min • in parallel with Track 1

| PAPER ID | PAPER TITLE                                                                                                                                                                               | PRESENTATION TIME   |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| 148      | Solar irradiation forecasting using CNN-LSTM: A hybrid model                                                                                                                              | 10:45 AM - 11:00 AM |
| 180      | Multi-Modal Wavelet Scalogram Fusion for Deep CNN-Based Capacitance Estimation                                                                                                            | 11:00 AM - 11:15 AM |
| 205      | Performance Enhancement of a DC-DC Boost Converter Using TD3-Based Reinforcement Learning Control                                                                                         | 11:15 AM - 11:30 AM |
| 259      | MetaphorMind: Cross-Lingual Metaphor Detector and Interpreter                                                                                                                             | 11:30 AM - 11:45 AM |
| 304      | Circuit Copilot: An AI-Based Assistant for Understanding Hand-Drawn Circuit Diagrams                                                                                                      | 11:45 AM - 12:00 PM |
| 322      | An Iterative Improved Framework for Severity Prediction and Aerosol Classification in Solar Photovoltaic Plants Using Integrated Deep Learning Fuzzy Logic and Predictive Control Process | 12:00 PM - 12:15 PM |

##### Track 1 - Renewable Energy & Storage - Session II (Parallel)

2:45 PM - 4:15 PM • 5 presentations × 15 min • in parallel with Track 2

| PAPER ID | PAPER TITLE                                                                                                   | PRESENTATION TIME |
|----------|---------------------------------------------------------------------------------------------------------------|-------------------|
| 470      | Comparative Analysis of LSTM based Day-Ahead Scheduling of Energy Storage Integrated Grid Connected Microgrid | 2:45 PM - 3:00 PM |
| 506      | A Novel SOC Based Control Algorithm for Enabling V2G and G2V Modes of Operation in EV Charging Station        | 3:00 PM - 3:15 PM |

#### IMPORTANT NOTICE TO AUTHORS & PRESENTERS

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**Track 1 - Renewable Energy & Storage - Session II (Parallel)**

2:45 PM - 4:15 PM • 5 presentations × 15 min • in parallel with Track 2

| PAPER ID | PAPER TITLE                                                                                                                                            | PRESENTATION TIME |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| 522      | Fast and Cost-Effective Cell Balancing for Lithium-Ion Batteries using a Single DC-DC Converter                                                        | 3:15 PM - 3:30 PM |
| 532      | Energy Management of a Solar/Wind/Second-life Battery Integrated System with SOH Estimation                                                            | 3:30 PM - 3:45 PM |
| 537      | Generation Scheduling of Power System with Thermal Power Plants, Electric Vehicles and Solar-powered Batteries with a Novel Cost function of Batteries | 3:45 PM - 4:00 PM |

**Track 2 - Computing, AI & Data Science - Session II (Parallel)**

2:45 PM - 4:30 PM • 6 presentations × 15 min • in parallel with Track 1

| PAPER ID | PAPER TITLE                                                                                                                           | PRESENTATION TIME |
|----------|---------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| 388      | Design of Power Efficient CNN Accelerator on FPGA                                                                                     | 2:45 PM - 3:00 PM |
| 449      | Comparative SOC Estimation for Li-Ion Batteries Using NN, 1D-CNN, and LSTM on Public Cycling Data                                     | 3:00 PM - 3:15 PM |
| 540      | Machine Learning-Based Reliability Improvement of a PSFB DC-DC Converter for Electric Vehicle Fast Charging                           | 3:15 PM - 3:30 PM |
| 742      | An OCPP Simulation Framework for Security Evaluation of EV Charging Infrastructure                                                    | 3:30 PM - 3:45 PM |
| 810      | Hybrid Kalman-Machine Learning Framework for Fault Detection and Isolation in Lithium-Ion Battery Systems                             | 3:45 PM - 4:00 PM |
| 821      | A Swarm Intelligence Assisted Neural Network based Maximum Power Extraction Strategy for Solar PV Applications with Resistive Loading | 4:00 PM - 4:15 PM |

**Day 2 - Saturday, 05 September 2026****Track 3 - Renewable Energy & Storage (Parallel)**

9:00 AM - 10:45 AM • 6 presentations × 15 min • in parallel with Track 4 and the PDW Panel Discussion

| PAPER ID | PAPER TITLE                                                                                                                   | PRESENTATION TIME   |
|----------|-------------------------------------------------------------------------------------------------------------------------------|---------------------|
| 669      | Advance Nuclear Energy System for Sustainable Transportation: Evaluating Nuclear Batteries as the Future of Electric Vehicles | 9:00 AM - 9:15 AM   |
| 794      | Defect-Tolerance Analysis of ZnO Nanorod Electron Transport Layers in Perovskite Solar Cells                                  | 9:15 AM - 9:30 AM   |
| 803      | Hybrid Optimization of Thermal and Battery Management for Enhanced EV Efficiency using ML Algorithms                          | 9:30 AM - 9:45 AM   |
| 809      | Field Data Analysis of Harmonic Distortion in Public PV-EV Charging Stations                                                  | 9:45 AM - 10:00 AM  |
| 819      | Optimization of Grid-Tied Photovoltaic Inverters Using TLBO, HHO, and Upgraded Addax Algorithm for Remote Areas               | 10:00 AM - 10:15 AM |
| 824      | Intelligent EV Battery Charging and Power Management in a DC Microgrid                                                        | 10:15 AM - 10:30 AM |

**Track 4 - Power Electronics & Control - Session I (Parallel)**

9:00 AM - 10:45 AM • 6 presentations × 15 min • in parallel with Track 3 and the PDW Panel Discussion

| PAPER ID | PAPER TITLE                                                                                                       | PRESENTATION TIME |
|----------|-------------------------------------------------------------------------------------------------------------------|-------------------|
| 823      | Next Generation Electrical Machines for Automotive Traction Applications Part I: 2-Wheeler and 3-Wheeler Vehicles | 9:00 AM - 9:15 AM |
| 281      | An Enhanced Control Strategy for Glitch-Free Mode Transition in Multi-Mode Power Electronic Converters            | 9:15 AM - 9:30 AM |

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**Track 4 - Power Electronics & Control - Session I (Parallel)**

9:00 AM - 10:45 AM • 6 presentations × 15 min • in parallel with Track 3 and the PDW Panel Discussion

| PAPER ID | PAPER TITLE                                                                                                                                                        | PRESENTATION TIME   |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| 370      | Hybrid AC/DC Transmission Architectures for Clean Energy Transition                                                                                                | 9:30 AM - 9:45 AM   |
| 451      | Review of Control Strategies in Mathematically Modelled Electric Vehicle Motors                                                                                    | 9:45 AM - 10:00 AM  |
| 697      | Mitigation of Power System Oscillations in Multimachine Networks During Transient States Using a Hybrid Fuzzy Logic and Neural Network-Based Power Flow Controller | 10:00 AM - 10:15 AM |
| 777      | Normalized Modeling and Optimal Design of SPS-Controlled Dual Active Bridge Converter                                                                              | 10:15 AM - 10:30 AM |

**Track 4 - Power Electronics & Control - Session II (Parallel)**

3:45 PM - 4:45 PM • 3 presentations × 15 min • in parallel with Track 5

| PAPER ID | PAPER TITLE                                                                                                                                                           | PRESENTATION TIME |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| 550      | Modelling and Simulation of an Intelligent Phase-Shift Controlled Isolated DC-DC Converter with Machine Learning-Based Fault Detection and Charging Mode Optimization | 3:45 PM - 4:00 PM |
| 528      | A Closed-Loop Control Strategy for Robust Operation of the Dual Active Semi-Bridge Converter                                                                          | 4:00 PM - 4:15 PM |
| 529      | Hybrid Covariance-Guided ABC-NSGA-II for Multi-Objective Transformer Design Optimization with Finite Element Analysis Validation                                      | 4:15 PM - 4:30 PM |

**Track 5 - Electronics & Embedded Systems (Parallel)**

3:45 PM - 5:00 PM • 4 presentations × 15 min • in parallel with Track 4 (Session II)

| PAPER ID | PAPER TITLE                                                                                                        | PRESENTATION TIME |
|----------|--------------------------------------------------------------------------------------------------------------------|-------------------|
| 199      | Impact of Model Complexity on Optimization-Based Control Strategies for Residential Battery Energy Storage Systems | 3:45 PM - 4:00 PM |
| 807      | Supercapacitor based TMS pulse generator design - Analysis of pulse shaping capability                             | 4:00 PM - 4:15 PM |
| 826      | Energy-Efficient JFET-Based Neuron-Like Signal Generator for Low-Power Neuromorphic Systems                        | 4:15 PM - 4:30 PM |
| 273      | Co-simulation of Real and Virtual ECUs for Left Shifting                                                           | 4:30 PM - 4:45 PM |

**Day 3 - Sunday, 06 September 2026****Track 7 - Emerging Tech & Interdisciplinary - Session I (Parallel)**

10:45 AM - 12:00 PM • 4 presentations × 15 min • in parallel with Track 6

| PAPER ID | PAPER TITLE                                                                                                                  | PRESENTATION TIME   |
|----------|------------------------------------------------------------------------------------------------------------------------------|---------------------|
| 184      | Remaining Life Time Estimation of Power Transformers                                                                         | 10:45 AM - 11:00 AM |
| 301      | Design and Implementation of an Autonomous Medical Observation & Remote Assistant                                            | 11:00 AM - 11:15 AM |
| 602      | Design and Simulation of a Robust Open-Loop Flyback Converter with an Intermediate Voltage Rail for Portable Medical Devices | 11:15 AM - 11:30 AM |
| 762      | Smart Grid Shield: IoT-Based Embedded System for Electricity Theft Detection                                                 | 11:30 AM - 11:45 AM |

**Track 6 - Power & Energy Systems - Session I (Parallel)**

10:45 AM - 12:45 PM • 7 presentations × 15 min • in parallel with Track 7

| PAPER ID | PAPER TITLE                                                                                                         | PRESENTATION TIME   |
|----------|---------------------------------------------------------------------------------------------------------------------|---------------------|
| 208      | Experimental Assessment of the Performance of a Rotationally Asymmetrical Concentrator Based Solar Window with Time | 10:45 AM - 11:00 AM |

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**Track 6 - Power & Energy Systems - Session I (Parallel)**

10:45 AM - 12:45 PM • 7 presentations × 15 min • in parallel with Track 7

| PAPER ID | PAPER TITLE                                                                                                                          | PRESENTATION TIME   |
|----------|--------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| 216      | Assessing the Societal Benefit of Hosting Capacity Flexibility: A Case Study of the Brazilian Market                                 | 11:00 AM - 11:15 AM |
| 274      | FPGA Based SPWM Controller for Five-Level Inverter                                                                                   | 11:15 AM - 11:30 AM |
| 789      | Control of a Three-Phase Grid-Connected Voltage Source Inverter Using a dq-Frame Current Estimator                                   | 11:30 AM - 11:45 AM |
| 806      | Comparative Evaluation of Front-End Converter Topologies for Supraharmonic Emission Reduction in Low-Voltage Grids                   | 11:45 AM - 12:00 PM |
| 227      | AI-Optimised Hybrid Flywheel-Hydrogen Fuel Cell System for Grid Frequency Stabilization in Sub-Saharan Power Networks                | 12:00 PM - 12:15 PM |
| 593      | Optimal Placement of Electric Vehicle Charging Stations with Distributed Generation in Distribution Networks: A Comprehensive Review | 12:15 PM - 12:30 PM |

**Track 7 - Emerging Tech & Interdisciplinary - Session II (Parallel)**

4:00 PM - 5:15 PM • 4 presentations × 15 min • in parallel with Track 6

| PAPER ID | PAPER TITLE                                                                                                                              | PRESENTATION TIME |
|----------|------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| 782      | Real-Time Vision-Based Multi-Zone Automatic Load Control Using Raspberry Pi for Indoor Energy Management                                 | 4:00 PM - 4:15 PM |
| 784      | Time-Dependent Degradation and Risk Evolution in Utility-Scale Photovoltaic Systems: Evidence From Long-Term Operational Data            | 4:15 PM - 4:30 PM |
| 827      | A Critical Analysis of Energy Harvesting and Intelligent Energy Management in Wireless Sensor Networks with a Value-Aware Decision Model | 4:30 PM - 4:45 PM |
| 387      | A Comprehensive Analysis of Cybersecurity Requirements for Electric Vehicles and Charging Infrastructure                                 | 4:45 PM - 5:00 PM |

**Track 6 - Power & Energy Systems - Session II (Parallel)**

4:00 PM - 5:30 PM • 5 presentations × 15 min • in parallel with Track 7

| PAPER ID | PAPER TITLE                                                                                                                           | PRESENTATION TIME |
|----------|---------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| 818      | Containerized SCADA Testbed for Supervisory Cybersecurity Analysis in Power System Infrastructure                                     | 4:00 PM - 4:15 PM |
| 610      | Reviving Legacy Brushless Exciter After Severe High Speed Rotating Rectifier System Damage of 120 MW Turbine Generator - A Case Study | 4:15 PM - 4:30 PM |
| 207      | Impact of the Reactive Power Control Modes of Wind Parks on the Power Systems Voltage Stability                                       | 4:30 PM - 4:45 PM |
| 531      | The role of synchronous condensers in power systems dominated by inverter-based resources                                             | 4:45 PM - 5:00 PM |
| 121      | Comparative Analysis of Heterogeneous Grid-Forming Inverters in Islanded Microgrid Operation                                          | 5:00 PM - 5:15 PM |

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